LEXSI LABS AI SAFETY & ALIGNMENT HACKATHON · TRACK 2b: REPAIRING SAFETY DRIFT WITHOUT RETRAINING

Same model, different language, different safety

Team: Aditya Arnav, Pallab Das · Mentor brief, 3 November 2026 · Core model: Tiny Aya Fire / Earth (3.35B), reference Tiny Aya Global · Repos: CuratorKIT, AuditKIT, CircuitKIT, SafeTune

The problem

Tiny Aya's regional models are tuned for the people who speak those languages: **Fire** for South Asia, **Earth** for Africa and West Asia. On average they are almost as safe as Tiny Aya Global, but in their **weakest language** they are 9–10 points less safe on MultiJail: 87.0% safe for Global, 78.1% for Fire, 77.5% for Earth (Tiny Aya tech report, Table 7). A harmful request Global refuses may be answered by the model built for that language's speakers, and a check on averages would not notice.

What we are doing

Step	What happens	Lexsi kit
1. Data	Harmful prompts (Aya red-teaming, MultiJail) and harmless look-alikes (XSTest) in English, Hindi, Bengali, Tamil, Hinglish, Arabic, Swahili. Splits are by item, so a translation of a test prompt can never be used for the repair.	CuratorKIT
2. Measure	Global, Fire and Earth, per language, on three axes: refusal of harmful prompts, wrongful refusal of harmless ones, and general ability (Global-MMLU-Lite). Two independent LLM judges plus a hand-checked sample, with and without Tiny Aya's default system preamble.	AuditKIT
3. Locate	EAP-IG finds the refusal circuit per language and per model, checked with the faithfulness tests. Where did regional tuning change it?	CircuitKIT
4. Repair	Four arms, compared head to head (below). Settings are tuned on the repair split only.	SafeTune
5. Verify	Re-test all three axes, then again after 4-bit compression (NF4 and GGUF Q4_K_M).	AuditKIT

The repair comparison

Arm	Edit	Why it is there
ReSta	Fire + α·(Global − Base), whole model	Established weight-space safety restore
Circuit (C-ΔΘ)	**The same safety vector, only in the layers CircuitKIT flags**	**Our main bet. Same edit as ReSta, so the difference isolates what the circuit map adds**
Merge-back	Blend Fire back toward Global	Naive baseline; likely costs regional ability
Steering	Refusal direction added at run time	Known to cause over-refusal; we check on Tiny Aya

All four are training-free weight or activation edits, matching the problem statement. A circuit-restricted LoRA is an optional fifth arm.

How we will judge success (pre-registered, before any repair)

- Close at least **half** of the worst-language refusal gap to Global.
- Wrongful refusal of harmless prompts rises by **at most 2 points** in any language; general ability falls by **at most 1 point**.
- All of the above **still holds at 4-bit**. Per-language 95% CIs and exact McNemar tests; every arm published, including losing ones, and negative results reported.

Where we are today

The private repository is set up. Data builds from the real sources, with the leakage check passing. Judges, statistics, the run ledger and all four kit integrations are written, and 28 tests pass. Everything runs on our own hardware (an Apple M1 Max with 64 GB, and free Kaggle T4s), with a free pair of 8B judges. The pre-registered 32B judges score the headline numbers, and we report how far the two pairs agree. Next: the day-1 smoke test (reproduce the gap on about 200 prompts, a first EAP-IG run on Tiny Aya), then the full baseline (4–5 Nov). We plan to contribute the kit gaps we found (English-only refusal detection, benchmark loaders, cross-language circuit tasks) upstream as pull requests.

What we would value your view on

- Do the success thresholds above look right to you?
- Is a public multilingual safety eval set (prompts and labels only, no harmful completions) acceptable under Cohere's acceptable-use policy?
- Have you seen pitfalls with refusal circuits or C-ΔΘ on Cohere2 models that we should plan for?